

Introduction to Microeconomics

Lecture 3—The Economic Method (continued): Some common pitfalls to avoid, more examples and seven important principles.

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Some Common Pitfalls for Decision Makers

Pitfall #1: Measuring Costs and Benefits as proportions rather than as absolute money amounts (like the Technodom example)

Pitfall #2: Ignoring Implicit Costs

If doing activity x means not being able to do activity y , then the value of you doing y is an **implicit cost** of doing x . You will end up doing bad decisions (and not behaving like a true economist!) if you forget the value of forgone opportunities. Thus, questions like

“Should I do x ?

should always be translated into ones like

Should I do x instead of y ?

where y is the most highly valued alternative to doing x .

Should you go skiing today?

In an ideal world, the NU campus should be in Almaty, and preferably, close to Shymbulak. Suppose we are in this ideal world. From experience you can say that a day on the slopes is worth \$50 to you. The charge for the day is \$30 (transportation and lift ticket).

If you don't go skiing, you will work at your new job as research assistant for Prof. Alejandro. The job pays \$40 per day, and you like it so much that you would have been willing to do it for free.

Should you go skiing or stay and work as research assistant?

$$\begin{aligned} C(x) &= \$30 + \$40 = \$70 \\ B(x) &= \$50 \end{aligned} \quad \left. \vphantom{\begin{aligned} C(x) &= \$30 + \$40 = \$70 \\ B(x) &= \$50 \end{aligned}} \right\} B(x) < C(x) \Rightarrow \text{you shouldn't go skiing.}$$

Do psychic cost of doing the job matter?

Suppose now that instead of being a research assistant, which you love, you would instead have a job scraping plates at the NU Canteen, which pays the same \$40 a day. However, this job is so unpleasant to you that you would be unwilling to do it for less than \$25 a day. Assume your manager at the canteen permits you to take the day off whenever you want. What should you do?

$x = \text{Go skiing.}$

$$\begin{aligned} \textcircled{1} \quad B(x) &= \$50 + \$25 = \$75 \\ C(x) &= \$30 + \$40 = \$70 \end{aligned} \quad \Rightarrow \text{Go skiing}$$

$$\begin{aligned} \textcircled{2} \quad B(x) &= \$50 \\ C(x) &= \$30 + (\$40 - \$25) = \$45 \end{aligned} \quad \Rightarrow \text{Go skiing}$$

Is it fair to charge interest when lending a friend some money?

Suppose a friend lends you \$10,000, and her primary concern when deciding whether to charge interest is to decide if it is fair to do so. She could have put that same money in the bank, where it would have earned, say, 5% interest, or \$500 each year. If she charges you \$500 interest for each year the loan is out, she is merely recovering the opportunity cost of her money. If she didn't charge you any interest, it would be the same as making you a gift of \$500/yr.

Why do banks pay interest for savings deposits?

Suppose you owned a bank and someone deposited \$10,000 in it on January 1 without your having to pay him interest. You could then take the money and buy a productive asset, such as a stand of trees. Suppose that each year trees grow at the rate of 6%, and that the price of a tree is proportional to the amount of lumber in it. At the end of the year you could then sell the trees for \$10,600, and have \$600 more than before. But that same option was available to the person who put his money in your bank. Why should he give you the \$600 he could have earned? He will be willing to let you use his money only if you compensate him for the opportunity cost of not using it himself. If you pay him 5% interest, he will get \$500, which will probably be acceptable to him because he won't have to go to the trouble of tending the trees himself (or of lending the money to someone who will tend them). You get to keep the remaining \$100 for taking care of that.

Failure to ignore sunk costs.

An opportunity cost will often not seem like a relevant cost when in reality it is. Another pitfall in decision making is that sometimes an expenditure will seem like a relevant cost when in reality it is not. Such is often the case with **sunk costs**, costs that are beyond recovery at the moment a decision is made. Unlike opportunity costs, these costs should be ignored

Should I drive to Pavlodar?

YOu are planning a 400 KM trip to Pavlodar (from Astana). Except for the cost, you are indifferent between taking the bus or driving. Busfare is \$100. You don't know how much it would cost to drive your car, so you call the local rental agency for an estimate. The attendant tells you that for your make of car the costs of a typical 16,000 KM driving year are as follows:

Insurance \$1000

Interest \$2000

Fuel and oil \$1000 These costs come to \$0.3/kilometer, and you use this figure to

Maintenance \$1000

Total \$5000

compute that the 400 KM trip to Pavlodar will cost you \$120. Since this is more than the busfare, you decide to take the bus.

Was the previous calculation correct?

No! Because you committed the error of not ignoring sunk costs! The insurance and interest payments do not vary with the number of kilometers you drive each year. Both are **sunk costs** and will be the same whether you drive to Pavlodar or not. Of the costs listed, only fuel & oil and maintenance are the only ones that vary with the miles driven. These total \$2000, or if you drive 16,000 kilometers yearly, \$0.125/kilometer. Thus, it would cost you to drive to Pavlodar \$50, since this is much less than the busfare, you should drive.

Pitfall #4: Failure to understand the average-marginal distinction

Suppose you own a fishing fleet consisting of a given number of boats, and can send your boats in whatever numbers you wish to either of two ends of an extremely wide lake, east or west. Under your current allocation of boats, the ones fishing at the east end return daily with 100 kilograms of fish each, while those in the west return daily with 120 kilograms each. The fish populations at each end of the lake are completely independent, and your current yields can be sustained indefinitely. True or False: If you shift some of your boats from the east end to the west end, you will catch more fish.

Fishing example: more data

In the fishing fleet scenario just described, suppose the relationship between the number of boats sent to each end and the number of pounds caught per boat is as summarized in the table. Suppose further that you have four boats in your fleet, and that two currently fish the east end while the other two fish the west end. Should you move one of your boats from the east end to the west end?

Number of boats	East end	West end
1	100 kg/boat	130 kg/boat
2	100 kg/boat	120 kg/boat
3	100 kg/boat	110 kg/boat
4	100 kg/boat	100 kg/boat

Output with 1 east, 3 west=430 kg.

$$\hookrightarrow 100 + 3(110) = 430$$

Current output (2 east, 2 west)=440 kg.

$$= 100 + 100 + 120 + 120$$

Marginal Benefit (West end)

$$1 \quad 130$$

$$2 \quad 240 - 130 = 110$$

$$3 \quad 330 - 240 = 90$$

How should you allocate resources?

The general rule for allocating a resource efficiently across different production activities is to allocate each unit of the resource to the productive activity where its marginal benefit is highest.

For a resource that is perfectly divisible, and for activities for which the marginal product of the resource is not always higher in one than in the others, the rule is: allocate the resource so that its marginal benefit is the same in every activity.

Similar fishing example

Number of boats	East end	West end
1	100 kg/boat	120 kg/boat
2	100 kg/boat	120 kg/boat
3	100 kg/boat	120 kg/boat
4	100 kg/boat	120 kg/boat

$\therefore$ 4 boats to the west end

Seven important principles

- ① The scarcity principle
- ② The cost-benefit principle
- ③ The incentive principle
- ④ The principle of comparative advantage.
- ⑤ The principle of increasing opportunity cost.
- ⑥ The equilibrium principle.
- ⑦ The efficiency principle.